Universal gravitation

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Predict how gravitational force changes with mass and separation.

SUCCESS CRITERIA - I can...

1. Use center-to-center separation.
2. Apply direct mass and inverse-square distance comparisons.
3. Identify equal forces on the two interacting bodies.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Bring back an earlier idea

A rover travels around a bend at a steady 5 m/s. Describe an observation that shows velocity changes even though the speed reading does not.

Marzano check - circle: ① ② ③

02 Explain a worked example

Reference: $F = Gm_1m_2/r^2$.
r is center-to-center separation. At fixed masses, doubling r makes the attractive force one quarter as large.

Worked reasoning: If separation doubles while both masses stay fixed, gravitational force becomes $1/2^2=1/4$ as large. A 20 N interaction would become 5 N.

Your turn: Sketch the two separations using centers. Draw equal and opposite force arrows for each pair.

Marzano check - circle: ① ② ③

03 Look and explain

Two equal spheres are shown. Mark the distance r used by the gravity equation and draw a force on each sphere.



Each circle represents one of the labeled objects or charges; r is their center-to-center separation.

Marzano check - circle: ① ② ③

04 Build the solution

The original force is 12 N. Predict it when one mass doubles, then separately when distance doubles.

Marzano check - circle: ① ② ③

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05 Use the idea

Does a large planet exert more force on a small moon than the moon exerts on the planet? Compare forces and accelerations.

Marzano check - circle: 1 2 3

06 Choose a method

A runner pushes the ground backward with 60 N. Name the partner force, its size and the object it acts on. Explain its role in the runner’s motion.

Marzano check - circle: 1 2 3

07 Exit check

At separation r the force is 8 N. At 2 r with masses unchanged, what is it?

Marzano check - circle: 1 2 3

08 STRETCH: Apply it another way

Starting from a force of 12 N, double one mass and double the separation. Find the new force and explain both effects separately.

Marzano check - circle: 1 2 3